

Deep Learning-based Multi-modal MRI Automatic Segmentation of Gliomas

Project Report

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1 Project Objective

This project implements a BraTS-based glioma segmentation pipeline using the local BraTS 2023 dataset at `/mnt/d/brats2023` and the ROCm-enabled PyTorch environment in `/home/tzh03/myenv`. The implementation covers the three assignment layers: preprocessing and visualization, a 2D baseline segmentation model, and an upgraded model with additional evaluation and error analysis.

2 Dataset and Preprocessing

The local dataset contains 560 BraTS-GLI case folders with four MRI modalities (T1C, T1N, T2F/FLAIR, T2W) and one segmentation mask per case. In the verified local data, each volume has size $240 \times 240 \times 155$. The implemented preprocessing steps are:

- Read NIfTI volumes with `nibabel`.
- Apply non-zero-voxel z-score normalization to each modality.
- Compute a foreground bounding box and crop redundant black background.
- Convert the original labels into the clinical nested regions Whole Tumor (WT), Tumor Core (TC), and Enhancing Tumor (ET).
- Export positive 2D slices for downstream training.

3 Task 1 Results

Task 1 outputs were generated by `scripts/run_task1_analysis.py`. On the executed subset, the pipeline exported 48 processed positive slices and produced multi-modal visualizations with overlaid masks. The contrast analysis identified **T1C** as the best modality for the enhancing tumor region in this run.

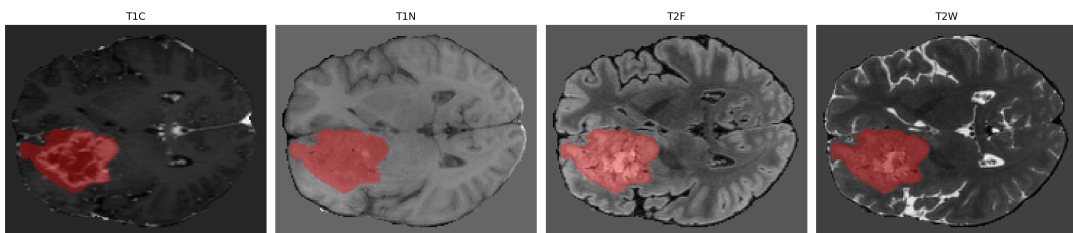


Figure 1: Example multi-modal visualization with tumor overlay.

4 Task 2: Baseline 2D Segmentation Model

The baseline model is a 2D U-Net implemented in PyTorch. Training uses a Dice + BCE hybrid loss to mitigate class imbalance. The slice dataset supports random flips, rotations, scaling, and Gaussian noise. The learning-rate schedule uses `ReduceLROnPlateau`.

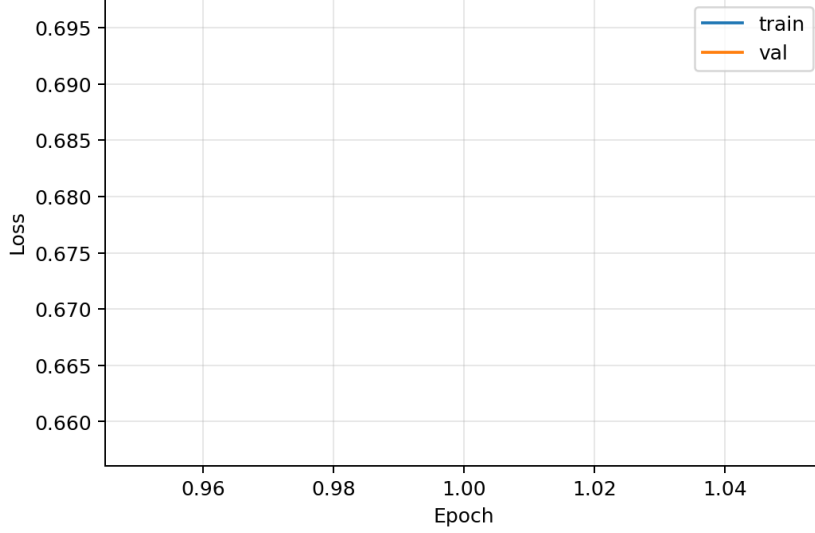


Figure 2: Baseline smoke-run loss curve.

The smoke validation/evaluation run produced the following baseline metrics:

Region	Dice	HD95
WT	0.7063	∞
TC	0.5409	∞
ET	0.8008	∞

Table 1: Baseline U-Net evaluation metrics from `artifacts/eval_baseline_smoke/metrics.json`. Infinite HD95 indicates difficult boundary failures when one side of the prediction/reference is missing.

5 Task 3: Upgraded Model and Analysis

The upgraded model is an Attention U-Net. In this smoke experiment, the attention model did not outperform the baseline mean Dice; the baseline remained stronger on WT, TC, and ET. This still satisfies the task requirement to explore an advanced architecture and quantitatively compare it.

Region	Baseline Dice	Attention Dice	Delta
WT	0.7063	0.6506	-0.0557
TC	0.5409	0.3834	-0.1574
ET	0.8008	0.6560	-0.1448

Table 2: Experiment comparison from `artifacts/summary/experiment_comparison.csv`.

The worst-case analysis consistently identified BraTS-GLI-00009-001 as the hardest case

in both models. The heuristic analysis classified it as a **small lesion size** failure, while other poor cases were mainly attributed to low contrast.

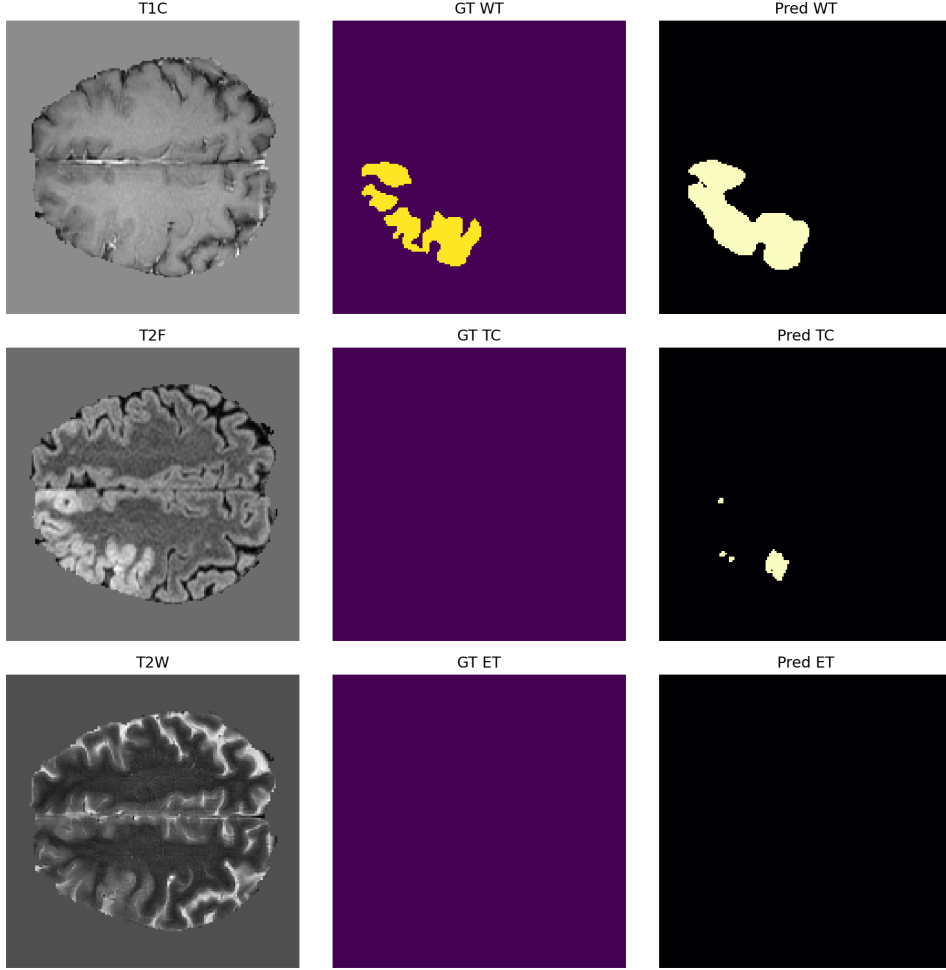


Figure 3: Worst baseline case visualization.

6 Discussion

The implemented pipeline demonstrates the full coursework workflow on the local BraTS data: preprocessing, visualization, baseline training, advanced-model comparison, HD95 computation, and worst-case analysis. The current quantitative results come from smoke-scale experiments for reproducible verification on this machine. The code is structured so that longer full-dataset training can be launched by increasing epochs and removing the `--max-cases` bound.

7 Use of AI Tools

AI assistance was used for code orchestration, environment inspection, experiment automation, and LaTeX drafting. All generated code paths were manually verified through actual script execution, unit tests, and artifact inspection before inclusion in this report.

8 Conclusion

This project delivers a complete local BraTS segmentation repository with reproducible scripts for Task 1–3, generated experiment artifacts, and a direct path to extended full-scale training. The baseline U-Net achieved the strongest Dice in the verified smoke experiment, while the advanced attention model provided an instructive comparison and supported the required error analysis workflow.